Modeling long-term vegetation recovery in the Okefenokee wetlands using remote sensing and generalized additive mixed models.

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INTRODUCTION

- The Okefenokee is a peat-dominated wetland in South Georgia and is one of the world's largest intact peat system (~150,319 ha; USFWS 2022)
- Fire is a natural ecological driver and influences wading birds and the endangered flatwoods salamander's habitat (Venne & Frederick, 2012; USFWS, 2021).
- Major recent fires in the Okefenokee include the 2007 Georgia Bay Complex (~300,000 ha) and the 2017 West Mims Fire (~59,000 ha).
- Post-fire vegetation recovery depends on burn severity, hydroperiod, soil properties, and vegetation type which have been seen in Argentina wetlands and the Everglades (Salvia et al., 2012; Bruland & Richardson, 2009).

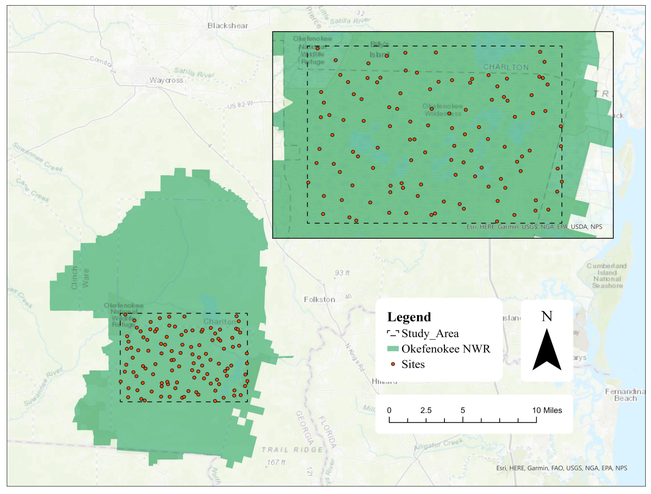

Figure 1. Study area within Okefenokee National Wildlife Refuge.

OBJECTIVES

There has been no published study that used long-term data to quantify multi-decadal vegetation trajectories in the Okefenokee. We address this gap by:

(i) To quantify fire severity for 114 sites from 2004 to 2024 with differenced Normalized Burn Ratio (dNBR)

(ii) to Model vegetation recovery using peak-season NDVI, EVI, and NDMI time-series in generalized additive models (GAMMs).

METHODS

Data Collection: Google Earth Engine

- Fire Detection: MODIS Burned Area
- Imagery: Landsat 5, 7, 8
- Site Selection (n = 100)
  - Grid-based sampling
  - ≥ 90 m from waterbodies
- dNBR Severity Classes (Key & Benson, 2006):
  - Low (0.10–0.27)
  - Moderate (0.27–0.44)
  - High (>0.44)
- Vegetation Indices (Used for recovery trajectories):
  - NDVI, NDMI, EVI

R Modeling

- Modeled nonlinear vegetation recovery trajectories
- Predictors:
  - dNBR class
  - Pre-fire NDVI, NDMI, EVI
  - Elevation
  - LANDFIRE vegetation type
  - Cumulative dNBR
  - Time since fire

Model Evaluation

- RMSE
- MAE
- R^2
- Leave-One-Site-Out (LOSO) Cross-Validation

RESULTS

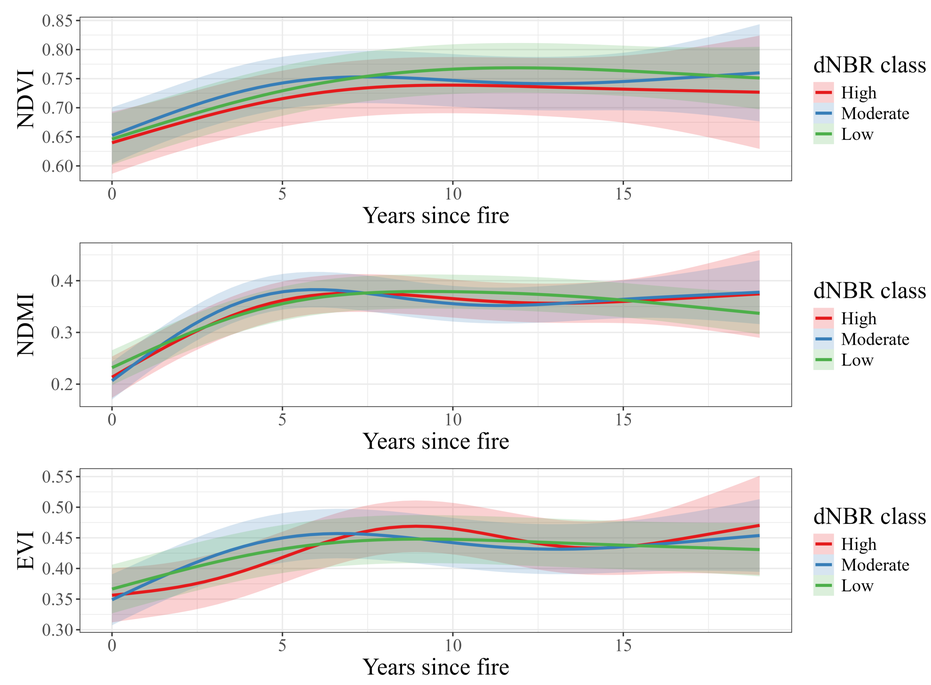

Figure 2. Modeled post-fire vegetation recovery for NDVI, NDMI, and EVI.

CONCLUSION

Table 1. Leave-One-Site-Out Cross Validation Metrics.

Metric	NDVI	NDMI	EVI
Root-mean-square error (RMSE)	0.1013	0.0615	0.0559
Mean absolute error (MAE)	0.0364	0.0446	0.0421
Coefficient of determination (RÂ²)	0.4978	0.6053	0.5751

Table 2. Model Fit Statistics for Full Dataset.

Metric	NDVI	NDMI	EVI
Root Mean Squared Error (RMSE)	0.106	0.0666	0.0627
Mean Absolute Error (MAE)	0.0391	0.0478	0.0474
Explained Variance (RÂ²)	0.0858	0.3328	0.2103

- Recovery trajectories were multi-phase, nonlinear patterns which aligned with peatland ecology. Most fires in the Okefenokee were low severity; high severity was rare.
- NDMI: best overall performance and generalizability.
- NDVI: fastest early recovery but least reliable.
- EVI: intermediate performance.
- Fire severity indicated as the strongest recovery driver and Cumulative dNBR indicated as strong disturbance indicator.
- Results support landscape-scale monitoring framework and useful for adaptive wetland and fire management.
- Future studies can include:
  - Dynamic vegetation maps
  - Hydrological data
  - High resolution fire detection method

REFERENCES

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